

cyd-dashboard — User Guide

A small self-contained touchscreen dashboard for the **Cheap Yellow Display** (ESP32-2432S028R). Once it's set up it runs on your WiFi — no computer needed.

What it shows

- **Clock & date** — always on top.
- **Weather** — current temperature, feels-like, humidity, sunrise/sunset, a 7-day forecast, and temperature history graphs (Day / Week / Month / Year).
- **Flights overhead** (*optional — on by default*) — a mini radar of live aircraft near you: callsign, route, altitude, speed, distance, airline logo, and the plane's actual ground track.
- **Pool temperature** (*optional — off by default*) — current water temperature and history graphs, if you have a Govee pool thermometer.
- **Sleep mode** — the screen sleeps overnight and wakes on touch.
- **Self-updates** — new firmware installs itself over WiFi.



First-time setup

1. **Plug it in.** On a brand-new device (or after a Factory Reset) it first asks you to tap a few crosshairs to calibrate the touchscreen — this is skipped if calibration is already saved.
2. **Enter WiFi.** Type your network name and password on the on-screen keyboard. (Change it later under **Settings** → **Network**.)
3. Done — the dashboard appears and starts loading weather and flights.

Set your location

Weather and flights need to know where you are. On first boot the device guesses your location from your internet connection — check that it's right:

Open **Settings** → **Location**, then use whichever is easiest:

- **Search Address** — type your city or address and pick the match.
- **Set** — type your latitude and longitude directly.
- **Find by IP** — guess again from your connection.

OpenSky API key (recommended)

Flights work out of the box, but a free OpenSky account raises your daily limit and removes the yellow "anonymous" warning.

1. Create a free account at opensky-network.org.
2. Go to **My OpenSky** → **Account** and create an **API client**. You'll get a **client ID** and a **client secret**.
3. On the device: open **Settings** → **Flight Tracker**, page right to the **OpenSky Credentials** screen, and enter both.

The three small readouts in the header — **CRP**, **CRL**, **CFT** — show your remaining daily OpenSky credits (tap them for detail). Grey is healthy, yellow is low, pink is nearly out. Credits reset daily.

Govee pool thermometer (optional)

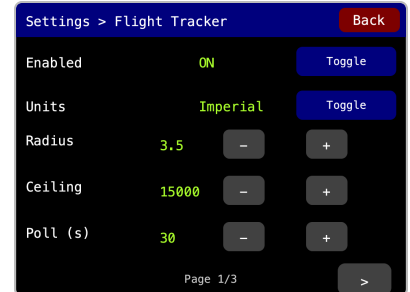
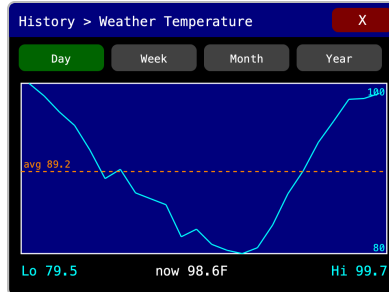
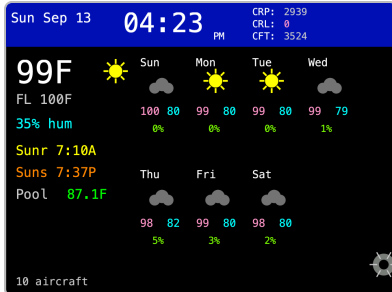
Only if you own a Govee pool thermometer:

1. Create a free developer account at developer.govee.com and generate an **API key**.
2. On the device: open **Settings** → **Pool Temp**, turn it **on**, enter the key, tap **Fetch Devices**, and pick your thermometer.

Using the dashboard

- **Cog (bottom-right)** — open settings.
- **Weather temperature (top-left)** — weather history graph.
- **Pool reading** — pool temperature history graph.
- **Aircraft count (bottom-left)** — details of the last overhead flight.
- **CRP / CRL / CFT (header)** — OpenSky credits detail.
- Any settings or graph screen returns to the dashboard after 2 minutes untouched.

The idle dashboard, the temperature graph, and Flight Tracker settings:



Settings at a glance

Defaults shown in parentheses.

- **General** — clock color (blue); auto-update (on).
- **Location** — your coordinates (auto-guessed on first boot).
- **Network** — WiFi network; DHCP or static IP; device hostname.
- **Flight Tracker** — enabled (on); units (imperial); radar radius (3.5 mi); altitude ceiling (15,000 ft); poll interval (30 s); timer bar (off); home airport (none); watched callsign (none); LED blink (on); Show IATA (on); OpenSky credentials.
- **Sleep Mode** — enabled (on); sleeps 10:00 PM – 8:00 AM; a touch wakes it for 5 minutes.
- **Pool Temp** — (off) enable, enter the API key, pick your thermometer.
- **Calibrate Touch** — rerun touch calibration.
- **About** — version, author, and firmware update status / install.
- **Reset** — see *Resetting* below.

Resetting (Settings → Reset)

Every option asks you to confirm and says exactly what it clears:

- **Restart** — reboots only; clears nothing.
- **Graph Data** — clears pool and weather temperature history.
- **Settings** — clears settings and credentials (keeps touch calibration).
- **Factory Reset** — clears *everything*: settings, credentials, files, history, and touch calibration. The next boot recalibrates touch and asks for WiFi again. **Warning:** if the device came pre-loaded with airline logos, they are erased — restoring them requires reloading them from a computer.
- **Cancel** — backs out without changing anything.

LED & border colors

- **Red border / LED** — something needs attention: no WiFi, bad OpenSky credentials, exhausted OpenSky credits, or missing weather/pool data.
- **Yellow** — OpenSky is running anonymously; flights still work.
- **Flight blinks** — blue for any overhead flight; red when it's departing your home airport; green when arriving; yellow when both ends are your home airport; white for your watched callsign.

Updates

The device updates itself over WiFi. **Auto-Update** (on by default) checks once a day; or open **Settings → About** and tap **Install** when a newer version is offered. If an update fails to start, it rolls back automatically.

Troubleshooting

- **Taps land in the wrong spot** — press and hold anywhere on the screen for **10 seconds** to recalibrate.
- **Wrong weather or no flights nearby** — fix your location under **Settings → Location**.
- **New WiFi network** — re-enter it under **Settings → Network**.